Ontology Design for Group Aware Summarization of McDonald's, Obama and Super Bowl Tweets

INFOIAG Intermediate Report 1

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ABSTRACT

This report presents an ontology design for group-aware summarization of tweets concerning McDonald's, Barack Obama, and the Super Bowl. The ontology combines five required domains (Person, Company, Country, Sport, and Health) with Advertisement and Claim as additional concepts. ClaimAssessment represents source-specific evaluations of claims. The design uses class hierarchies, property characteristics, and domain constraints to support entity identification and consistency checking. We describe two summarization scenarios and examine how compression, dominant viewpoints, and incorrect attribution can affect the resulting summaries. We also specify ontology queries and a validation procedure for examining inferred relationships, inconsistent assertions, and the connection between source viewpoints and summary content. The report distinguishes implemented and tested components from proposed extensions and remaining work.

KEYWORDS

Ontology, LLM Agents, DivSumm, Tweet Summarization, Bias Detection, Knowledge Representation

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1 TASK AND SELECTED TOPICS

1.1 Task

The goal of this project is to develop an ontology that can support an LLM-based agent in summarizing collections of tweets. The ontology provides structured knowledge about the entities and relations that occur in the data. This provides the agent with structured information that can be used to check the content of generated summaries.

The ontology covers five main domains: Person, Company, Country, Sport and Health. In addition, Advertisement and Claim are included to represent information that is especially relevant to the selected topics. The ontology is populated with instances and relations that can later be queried during the summarization process.

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1.2 Selected Topics

Three topics were selected: McDonald's, Barack Obama and the Super Bowl. They represent different types of entities and thus require different parts of the ontology. McDonald's typically concerns a company and its products, Barack Obama concerns a person and political claims, while the Super Bowl concerns a sporting competition and related events.

The same ontology can thus be used across all three tweet collections. Topic-specific information is represented as individuals within this shared structure. For example, McDonald's is represented as an individual of the `Company` class, Obama as an individual of `Person`, and the Super Bowl as an individual of `Competition`.

1.3 Dataset and Report Scope

We use the DivSumm dataset introduced by Olabisi et al. [1], which contains 2,250 tweets across 25 topics. Each topic contains 90 tweets, with 30 tweets from each of three dialect groups: White-aligned, Hispanic, and African-American. Our selected topics, McDonald's, Barack Obama, and the Super Bowl, therefore comprise 270 input tweets. Each topic also has two human-written abstractive reference summaries of five sentences each.

The equal number of input tweets per group allows us to examine whether summarization disproportionately represents particular groups despite their equal input sizes. Equal input representation does not imply that each group expresses a single viewpoint or that a fair summary must allocate exactly the same number of sentences to each group. We use the supplied labels as dataset annotations rather than verified demographic identities of the authors.

This intermediate report focuses on ontology design, two realistic summarization prompts, the expected use of ontology reasoning, and ontology queries. The completed agent and comparative evaluation belong to later project stages.

2 ONTOLOGY DESIGN

The ontology includes five main domains: Person, Company, Country, Sport, and Health. We also include two additional concepts: Advertisement and Claim. The same ontology schema is used across the three selected DivSumm topics: McDonald's, Barack Obama, and the Super Bowl. Instead of constructing separate ontologies for each topic, we represent topic-specific entities as instances of these shared classes.

2.1 Shared Model and Topic-Specific Content

The five required domains each have their own class hierarchy, but they are still supported by additional classes to allow relations between otherwise isolated parts of the ontology. For example, the

Person domain uses classes such as Occupation, Role and Team; Company uses Industry, Product and Branch; Country uses City, Landmark and Language; and Sport uses Match, Competition and Outcome. The Health domain contains classes such as Disease and Symptom, with additional supporting classes such as Food, FoodCategory and Region. The vocabulary is similarly reused across the McDonald's, Barack Obama and Super Bowl topics, while the topic-specific information is represented through different instances.

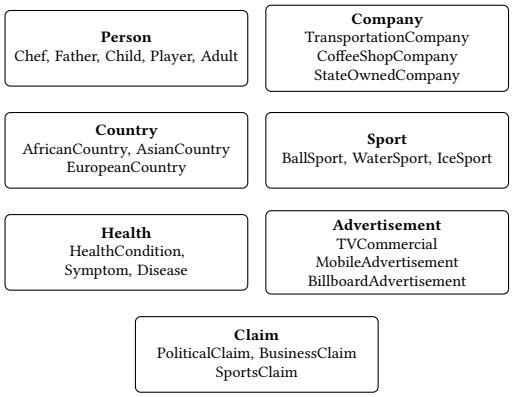


Figure 1: High-level overview of the ontology class hierarchy. Person, Company, Country, Sport, and Health form the five main domains, while Advertisement and Claim are the two additional concepts.

Table 1: Supporting classes used by the main ontology domains.

Area	Supporting classes
Person/social	Occupation, CharacterTrait, Role, Team
Company	Industry, Product, Service, Branch, ResearchAndDevelopment
Country/geography	City, Landmark, Food, Language, GovernmentForm, Holiday, Event, Currency, InternationalOrganization
Sport	Match, Competition, League, Tournament, Outcome, Win, Loss, Draw, Ranking, MatchResult, Rule, Position, SkillLevel, WinnerDetermination, Ritual
Health	Food, FoodCategory, Region
Advertisement	Location, AgeGroup
Claim	ClaimAssessment, Source, Event

The shared ontology is populated with topic-specific individuals. In Table 2 we present the representative instance relations for the three selected topics.

Table 2: Representative instance relations in the ontology.

Topic	Representative instance relations
McDonald's	`McDonalds headquarteredIn America`; `McDonalds produces FastFoodProduct`; `McDonalds operatesIn FastFoodIndustry`; `McDonalds employs chef1`; `McDonalds employs waitress1`; `McDonalds partnersWith OtherCompany`; `McDonalds investsIn R&D_Project1`; `McDonalds hasBranch branch1`.
Obama	`Obama hasCountryOfOrigin America`; `Obama hasOccupation president`; `Obama hasCharacterTrait Rational`; `Obama friendsWith Person1`.
Super Bowl	`SuperBowl1 hasRitual HalftimeShow`; `SuperBowlMatch1 partOf SuperBowl1`; `SuperBowlMatch1 hasParticipantTeam team1`; `SuperBowlMatch1 hasParticipantTeam team2`; `SuperBowlMatch1 hasWon team1`; `SuperBowlMatch1 hasLost team2`.

2.2 Additional Concepts and Their Purpose

The ontology extends the five main domains with *Advertisement* and *Claim*. These concepts represent information relevant to the selected topics that is not captured by the five existing domains above alone.

Advertisement. An Advertisement represents promotional content rather than the company or product itself. Modelling advertisements as separate entities therefore helps distinguish reactions to promotional material from broader opinions about McDonald's, its products, or the Super Bowl. To represent this distinction, the ontology links each Advertisement to the company responsible for it and the product being promoted. Additional properties capture where the advertisement appears and which audience it targets.

Claim. A Claim represents a statement concerning a person, company, sport, location, or event. Claims are divided into PoliticalClaim, BusinessClaim, and SportsClaim.

ClaimAssessment. A ClaimAssessment represents a source's evaluation of a claim and records whether the claim is judged to be valid.

Table 3: Structure of the two additional concepts.

Concept	Relation / property	Target or role
Advertisement	`createdBy`	Company that created the advertisement
	`promotes`	Promoted Product
	`shownAt`	Location where it is shown
	`hasLanguage`	Language/version
	`targetsAgeGroup`	Intended AgeGroup
Claim	`madeBy`	Person making the claim
	`relatedTo`	Company, Person, Sport, or Event
	`aboutLocation`	Location concerned by the claim
ClaimAssessment	`assessmentOf`	Claim being assessed
	`assessedBy`	Source performing the assessment
	`isValid`	Source-specific Boolean validity value

2.3 Property Characteristics

The ontology includes at least three transitive, two symmetric, three functional, and two inverse-functional object properties. Each property characteristic is only used where the resulting OWL inferences are consistent with the intended meaning of the relation.

Table 4: Object-property characteristics selected for the ontology.

Characteristic	Property	Rationale
Transitive	`locatedIn`	Nested geographical containment
Transitive	`partOf`	Transitive part–whole relation
Transitive	`hasDescendant`	Propagates across generations; `hasChild` is not transitive
Symmetric	`friendsWith`	Friendship is reciprocal
Symmetric	`marriedTo`	Marriage is reciprocal
Functional	`hasOccupation`	At most one occupation in the current model
Functional	`hasCapital`	At most one capital per country
Functional	`headquarteredIn`	At most one headquarters country per company
Inverse Functional	`hasCapital`	At most one country per capital city
Inverse Functional	`hasHeadOfState`	Current assumption: one country per head-of-state individual

Inverse-functional properties should be used with caution since, if two individuals are linked to the same object through such a property, the reasoner may infer that those individuals are actually the same entity. Thus, these properties should only be used when sharing the same object is intended to indicate that the linked individuals are the same entity.

2.4 Ontology Constraints and Implementation Boundaries

In addition to the property characteristic described above, the design contains several domain-specific constraints. Some of them can be expressed directly through class restrictions or functional properties, while others still require formalisation before implementation and reasoner testing.

Table 5: Main ontology constraints and their current design status.

Area	Constraint	Status
Person age	`hasAge` ≤ 110	Designed
Adult age	Adult age ≥ 18	Designed
Marriage	`marriedTo` relates Adult individuals and is symmetric	Designed
Occupation	`hasOccupation` is functional	Designed
Character trait	Reserved $\sqsubseteq$ Introvert	Modelling choice
City–country	`belongsToCountry` is functional	Designed
Capital	`hasCapital` is functional	Designed
Head of state	`hasHeadOfState` is functional and inverse functional	Modelling assumption
Health	Allergy and Injury are disjoint	Current design
Player/team	At most one team within a league at a given time	Formal temporal encoding remains
Match outcome	At most one outcome	Functional `hasOutcome`; existence separate
Claim assessment	One Claim, one Source, and one Boolean validity value per assessment	Designed via ClaimAssessment

3 SUMMARIZATION PROMPTS AND EXPECTED REASONING

3.1 Prompt 1: General reader briefing

> "Below are 90 tweets about [TOPIC]. Write a 5-sentence summary for someone who has not followed this topic, so they get a quick picture of what people are saying."

Why this prompt may lead to biased summarization

Readers who want a "quick picture" might push the model toward the most frequent or extreme opinions. Viewpoints from less dominant dialects can become excluded in generalisations. For example, the mixed reaction to the fish-product advertisement (MCD_019) could disappear from a McDonald's summary.

3.2 Prompt 2: Communications analysis

> "I work in communications and need to decide how to respond to public reaction. Summarize the tweets about [TOPIC] in 5 sentences, highlighting the main sentiment and the most important complaints or praise."

Why this prompt may lead to biased summarization

The response to this prompt will guide a real decision, so the LLM may also weigh that in its response. Besides that, the wording "main sentiment" and "most important" invites generalising one group's opinion as that of "the public". This risks creating an unfair picture of what is being said, in favour of whichever group's sentiment the LLM deems "most important".

3.3 Expected agent behaviour and reasoning

(1) **Draft.** The agent first generates a five-sentence summary from the tweets.
(2) **Extraction.** It extracts the claims made in the tweets, each linked to its author (`madeBy`), the entity it concerns (`relatedTo`), and, where relevant, a location (`aboutLocation`). It does the same for every sentence of the draft summary.
(3) **Coverage check**. The agent queries the ontology to see whether claims from every dialect group on the main topics are represented in the draft. Any claim that no summary sentence represents is flagged as an omitted claim.
(4) **Target check**. The agent checks that each sentence's target entity (via `relatedTo`) matches the entity the source tweets are actually about. For example, praise for a Taco Bell advertisement (SB_060), linked via `promotes`, should not become praise for the Super Bowl competition in general.
(5) **Assessment check** (*Q5*). The agent checks whether each claim represented in the summary has a `ClaimAssessment`, and if so, what its `isValid` value is and which `Source` it is `assessedBy`. A claim with no assessment, or one assessed as invalid, is flagged for review.
(6) **Fact and consistency check** (*Q1-Q4, Q6*). The agent compares claims about entities with the facts in the ontology, using the Person, Company and Sport domains. It also checks that no two summary sentences contradict each other.
(7) **Revision and re-check.** The agent rewrites the flagged sentences, then runs the checks again on the revised summary before returning it.

Limits. The ontology does not contain all facts. When it has no information on a claim, the agent flags it as unverified instead of treating it as wrong. The aim is to signal likely bias and inconsistency, not to guarantee a correct summary.

4 ONTOLOGY QUERIES AND VALIDATION

4.1 Query Scenarios

When a draft summary makes a factual statement about McDonald's, Obama, or the Super Bowl, the agent can represent the statement as a `Claim`. Relevant ontology queries can then be used to check the statement against the available knowledge. The result can subsequently be represented through a `ClaimAssessment`.

Q1 – Employees (Company). Who works for McDonald's? The ontology contains the inverse properties `employs` and `worksFor`. This allows the relation to be queried from the perspective of the employee.

```
SELECT ?person WHERE {
  ?person :worksFor :McDonalds .
}
```

Result: `chef1`, `waitress1`.

Q2 – Partners (Company). Which companies are partners of McDonald's? The query retrieves companies connected to McDonald's through the `partnersWith` relation.

```
SELECT ?company WHERE {
  :McDonalds :partnersWith ?company .
}
```

Result: `OtherCompany`.

Q3 – Match Result (Sport). Who won and lost the Super Bowl match, and which ritual belongs to the competition containing the match?

```
SELECT ?winner ?loser ?ritual WHERE {
  :SuperBowlMatch1 :partOf ?competition ;
                   :hasWon ?winner ;
                   :hasLost ?loser .
  ?competition :hasRitual ?ritual .
}
```

Result: `team1` won, `team2` lost, and the associated ritual is `HalftimeShow`. A generated summary containing a different winner or loser can therefore be compared with these ontology facts.

Q4 – Person Facts (Person). What is Obama's occupation and which individuals are connected to him through `friendsWith`? Because `friendsWith` is symmetric, a relation can also be retrieved in the opposite direction from the one explicitly asserted.

```
SELECT ?occupation ?friend WHERE {
  :Obama :hasOccupation ?occupation .
  ?friend :friendsWith :Obama .
}
```

Result: `president`; `Person1`.

Q5 – Assessed Claims (Claim). Which claims have been assessed as false, and which source performed the assessment? A `ClaimAssessment` is designed to connect a claim to its source and validity value.

```
SELECT ?claim ?text ?source WHERE {
  ?assessment :assessmentOf ?claim ;
              :assessedBy ?source ;
              :isValid false .
  ?claim :hasText ?text .
}
```

Result: The current ontology does not yet contain instances of `ClaimAssessment`, so this query currently returns no results. These instances can be created when claims from generated summaries are evaluated.

Q6 – Contradiction Detection (Consistency). The ontology can also be used to detect inconsistent information. For example, `Allergy` and `Injury` are defined as disjoint classes. If the same individual is classified as both, the reasoner detects an inconsistency.

For example, adding the following assertions creates a contradiction:

```
:ExampleCondition a :Allergy .
:ExampleCondition a :Injury .
```

This provides a simple way to test whether information extracted from a generated summary conflicts with the constraints defined in the ontology.

4.2 Example Executable Query

The queries can be executed in Protégé using the reasoner and the SPARQL query interface. Q4 provides an example in which reasoning is required for the symmetric `friendsWith` property. If the ontology only contains

```
:Obama :friendsWith :Person1 .
```

the symmetric property allows the reasoner to derive

```
:Person1 :friendsWith :Obama .
```

The query can therefore retrieve `Person1` even when this second direction is not explicitly stored in the ontology.

REFERENCES

[1] Olubusayo Olabisi, Aaron Hudson, Antonie Jetter, and Ameeta Agrawal. 2022. Analyzing the Dialect Diversity in Multi-document Summaries. In *Proceedings of the 29th International Conference on Computational Linguistics*. 6208–6221.